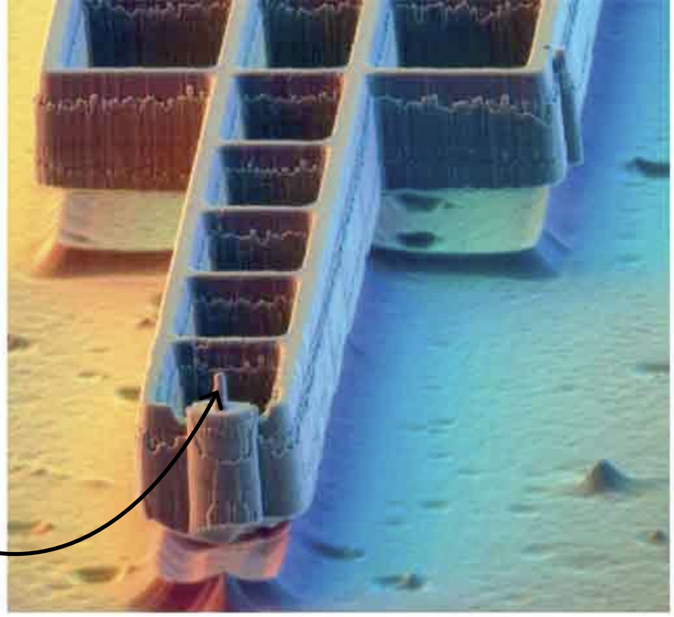
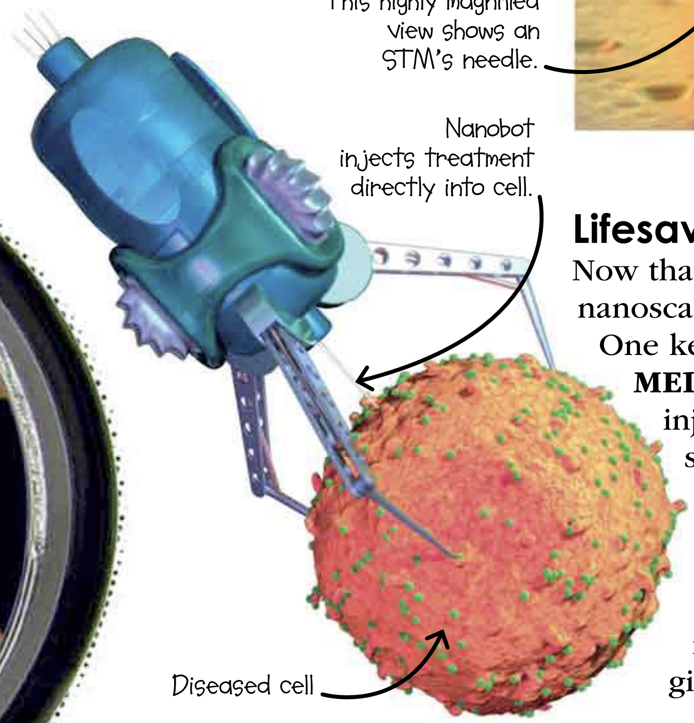
Marvelous microscope

In 1981, the **scanning tunneling microscope** (STM) was invented by German scientists **GERD BINNIG** and **HEINRICH ROHRER**. It uses a needle tip just a few atoms wide to scan an object, tracing out the surface atoms and spaces between them to form an image. STMs work at **incredible resolutions**, capable of ***showing us individual atoms***. They also allow scientists to work at the nanoscale directly, moving and manipulating individual atoms for the first time.



This highly magnified view shows an STM's needle.



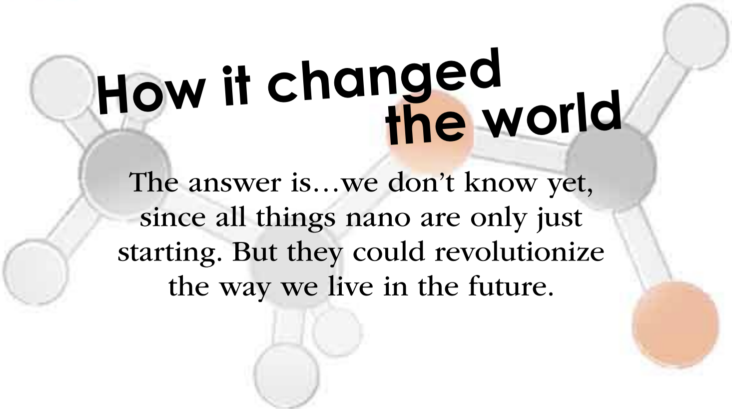
Nanobot injects treatment directly into cell.

Diseased cell

Lifesaving nanobots

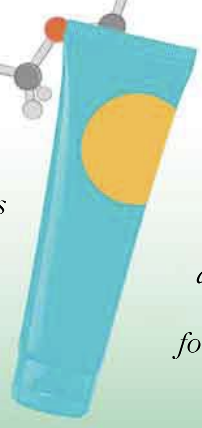
Now that scientists are able to work on the nanoscale, ***the possibilities are endless***. One key application in the future could be **MEDICAL NANOBOTS**—tiny robots injected into the body. Some might scrub blood vessels clean of fats while others could **repair damage** from the inside, or track, capture, and deal with harmful bacteria or diseased cells (left). ***Swarms of nanobots*** might monitor you from the inside to give your body a continual checkup.

Did you know?
These micro-cogs, shown here next to a fly's leg, were made using techniques that can work at scales of thousandths of a millimeter.

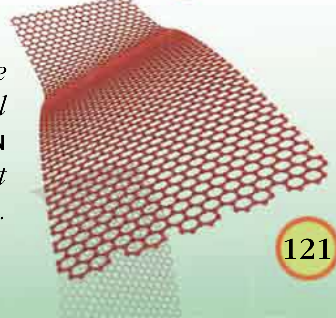


How it changed the world

The answer is…we don't know yet, since all things nano are only just starting. But they could revolutionize the way we live in the future.



NANOPARTICLES *of metal oxides are used in some* **SUNSCREENS**. *They offer protection from the sun but don't leave white marks on the skin.*



GRAPHENE *is a remarkable material with many potential applications. It is made of* **CARBON ATOMS** *joined in hexagons that form a surface a single atom thick.*